# MATH290

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**Proposition 1.** *For any integer $k$, $k^3 - k$ is a multiple of 6.*

*Proof.*

Firstly, note that the sum of two multiples of 6 is itself a multiple of 6. An integer is a multiple of 6 if it can be expressed as the product of 6 and some other integer. Consider a multiple of six $n = 6i$ and another $m = 6j$. Observe,

$$\begin{aligned} n + m &= 6i + 6j \\ &= 6(i + j) \end{aligned}$$

Relying on the property that summing integers yields an integer, we've shown that $n + m$ can be expressed as a product of 6 and some other integer $i + j$. That is, $n + m$ is a multiple of 6 for any multiples of six $n$ and $m$.

Secondly, observe what happens if you substitute $-k$ for $k$:

$$\begin{aligned} &(-k)^3 - (-k) \\ =& -(k^3) + k \\ =& -(k^3 - k) \end{aligned}$$

Thus, if $k^3 - k$ can be expressed as $6n$ for some integers $k$ and $n$, then $(-k)^3 - (-k)$ can be expressed as $-6n$. Therefore, if I can show that $k^3 - k$ is a multiple of 6 for all integers $k \geq 0$, then $k^3 - k$ is a multiple of 6 for all integers $k < 0$. With this in mind, I'll only focus on integers $k \geq 0$ for the rest of the proof.

Let $f_k$ be the sequence $k^3 - k$ for integers $k \geq 0$. It turns out that rather than proving each term in this sequence is a multiple of 6, it's easier to prove that *the difference between consecutive terms* is a multiple of 6. (Or at least *I* think it is.) Let's let $g_k$ be the difference between any $f_{k+1}$ and $f_k$. That is,

$$\begin{aligned} g_k &= f_{k+1} - f_k \\ f_{k+1} &= f_k + g_k \end{aligned}$$

If I can show that $g_k$ is a multiple of 6 for all integers $k$.

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